Robot Autonomy - Group Project

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Due Date: See submission section

Introduction

This is an asynchronous group project, which means that it is to be completed outside of lecture times and ultimately will involve running code on the robot. We encourage you to prepare as much as possible in simulation before you switch to the robot. This is also a group assignment, and the group should be the same as for the 2nd and 3rd labs. You only need to create one code base, video, and report submission per group.

The purpose of this lab is to give you the opportunity to explore robot autonomy further in a hands-on manner. We have covered a wide range of topics over the semester. Use this oppurtunity to dive deeper and explore a couple of topics in more depth. This assignment is also an opportunity to explore the utility of combining different approaches into a larger system.

Different levels of abstraction of tasks

NOTE - ROBOT EXECUTION AND SAFETY

Safety is paramount! Remember to always work with a partner (you are NOT authorized to work alone on the robots) and someone should have their hand on the emergency stop whenever the robot is running. Hit the stop immediately when anything happens that is even potentially close to unsafe for a human or the robot. Just because something works in simulation does NOT mean it will work exactly the same on the real system.

Robot Setup and Simulation

The robot will be setup in a shelf environment as with three shelves on its right and one on its left (robot's perspective), as shown in the second lab's simulation environment and the figure (shelf surfaces will be added this week). The shelves should not be modified. We will setup the physical systems as close as possible to the simulated one. Each robot will be shared between

Figure 1: The robot environment including 1 shelf on the right of the robot, 3 shelves on its left, and a clear table region in front of the robot

2-3 groups. We recommend using the same robot as there may be small yet critical differences between the setups. Each robot is also mounted differently relative to the shelves in the x-y plane, and the groups should take this shift into account. Feel free to share robot-specific information between groups.

Project Task

The goal of the project is for you to create your own task for the robot to complete autonomously. The scope of the project should be similar to the previous labs' tasks.

Example project task ideas include:

1. Picking up different items and rearranging them by weight
2. Picking up a stack of blocks and carrying them without dropping the top one.
3. Creating a wall of blocks. Use different colors of blocks to make pixel art.
4. Stocking the shelves with items.
5. Playing a simple game, e.g., placing blocks for tic-tac-toe.
6. Use a whiteboard pen to write on the surface.
7. Learning a DMP or diffusion policy for moving between locations on different shelves.

Most importantly - it should be fun and exciting for you!

Projects will generally focus on some aspects of the autonomy stack more than others (e.g. a project may focus more on task planning with little challenge in motion planning). Try to select a project that focuses on the areas that you and your team are the most excited to explore in more depth. Some more complex projects may explore multiple areas. To maintain a reasonable scope, we suggest including simple placeholders for parts of the project in order to first focus on completing others. You want to avoid projects where multiple parts need to all come together perfectly in the end or else nothing works. A few additional important points to keep in mind when selecting a task:

1. You may use the wooden blocks that we have used in the labs, but you are not limited to using blocks.
2. Items that pose a hazard to the robot (e.g., liquids, powders, hot items) or are difficult to clean up (e.g. beads) are not allowed near the robot.
3. You are not allowed to adjust or remove the shelving structure.
4. You should not remove the gripper from the robot.
5. We do not provide cameras or support for cameras. We recommend focusing on tasks with known object poses.
6. You may still incorporate a camera in the robots environment if you like (note: no wrist camera mounts). However, the setup and visual processing should only be a small part of your presentation and final report (e.g., 1-2 paragraphs).
7. No throwing/striking of items.

Submission

Presentation: You will be presenting your project in the last week of class (April 21st and April 23rd). The presentation should be seen as progress presentations, including both what you have achieved so far and what you still plan to achieve by the end of the following week. The presentation should be around 8min long to give 2-3min time for feedback from the rest of the class.

Final report: The final report is due on May 1st, no extensions. The report should be 2-3 pages long, including pictures, in two-column IEEE conference style, i.e., similar to an ICRA or IROS paper.

The presentation and final report should both include sections that:

1. Motivate the task problem and emphasize why it is interesting/challenging.
2. Describe the overall framework that you created
3. Describe the individual components
4. Describe how the work was divided among the team.
5. Present the results of the autonomous executions. What worked and what did not?

Final video: The final video is due on May 1st, no extensions. The final video should demonstrate the robot performing the task autonomously. The videos should be 1-3min long. You may add narration, title cards, etc. if you wish, but a basic video of the robot task would be sufficient. We will upload the final videos to canvas for others to see.

Disclaimer

This is the first time that we are running the group project in this manner. If you have any feedback or suggestions (e.g., the shelf heights should be adjusted for everyone for better performance, or there are issues with the simulation), or are experiencing any sources of frustration, please let us know as soon as possible on piazza. **The labs and the final project are meant to be fun!**

We are in the process of setting up the real robots and will let you know when they are ready for the real world evaluation.